CHIRANTHA K M

PYTHON ENGINEER · APPLIED AI/ML · ENTERPRISE DATA PLATFORMS · FULL-STACK

Bengaluru, India | +91 83102 28282 | chirantha.km15@gmail.com
github.com/chirantha-km | linkedin.com/in/chirantha-km | chirantha-km-developer-portfolio.vercel.app

SUMMARY

Python software and applied AI/ML engineer experienced in building production data pipelines, backend APIs, full-stack applications, and statistical ML solutions. Processed 50,000+ financial records daily with 99.4% reliability; improved SQL reconciliation query performance ~10×. Experienced in end-to-end technical delivery, observability, and cross-functional collaboration; researching LLM output validation and hallucination detection.

TECHNICAL SKILLS

Programming & Data Python, OOP, data structures, Pandas, NumPy, data ingestion, transformation, validation

Applications & APIs FastAPI, REST APIs, full-stack web/mobile applications, microservices, integrations, JWT, RBAC

Enterprise Data PostgreSQL, SQL, financial data, reconciliation, query optimization, indexing, partitioning, schema design

Applied AI/ML Scikit-learn, XGBoost, anomaly detection, feature engineering, model evaluation; LLM reliability research

Cloud & Delivery Git, pytest, integration testing, CI/CD, GitHub Actions, Docker, Kubernetes, AWS (working knowledge), GCP

Streaming & Operations Kafka, Redis, event-driven pipelines, Prometheus, Grafana, logging, reliability, retry logic

EXPERIENCE

Reverb Inc. *Remote*
Quant Analyst & Tech May 2026 – Sep 2026

- Translated ambiguous quantitative business questions into data workflows, experiments, and implementable technical solutions with stakeholders.
- Owned data preparation, analysis, debugging, and end-to-end delivery; iterated on solutions based on results and business needs.

MSR Solutions *Bengaluru, India*
Full-Stack Engineer & Technical Lead (Intern) Dec 2025 – May 2026

- Sole engineer shipping production web, mobile, admin, and CRM applications; designed backend APIs, integrations, and maintainable services.
- Built an ML-powered estimation feature using cost and timeline models; owned architecture, implementation, testing, and production deployment.

SuperBuilt AI *Remote*
Backend / Data Engineer Intern Feb 2025 – May 2025

- Engineered Python/SQL financial data pipelines handling 50,000+ records daily; implemented validation, reconciliation, retries, and dead-letter routing for 99.4% reliability.
- Optimized PostgreSQL reconciliation queries using composite indexes and partitioning, cutting latency from 8s to 900ms (~10×).
- Built statistical anomaly detection using Z-score and Isolation Forest; added Prometheus observability and resolved data-quality failures.

YBI Foundation *Bengaluru, India*
Data Science Intern Jan 2024 – Feb 2024

- Built reproducible Python ML pipelines for ingestion, feature engineering, training, and evaluation with 78% pytest coverage.
- Reduced debugging MTTR from 4 hours to under 45 minutes with structured logging and Grafana dashboards.

SELECTED DATA, AI/ML & SOFTWARE PROJECTS

Real-Time Statistical Anomaly Detection Engine

PYTHON · FASTAPI · KAFKA · SCIKIT-LEARN · POSTGRESQL · PROMETHEUS · GRAFANA · DOCKER

- Processed 500 transactions/sec through Kafka; combined Z-score, IQR, and Isolation Forest with PostgreSQL persistence, API endpoints, and latency monitoring.

Production REST API — Task & Team Management

FASTAPI · POSTGRESQL · REDIS · JWT · DOCKER · PYTEST · GITHUB ACTIONS

- Shipped tested REST APIs with JWT/RBAC and Redis rate limiting; achieved sub-100ms p95 latency under 500 concurrent users and 84% pytest coverage.

Real-Time Event Streaming & Notification Service

PYTHON · FASTAPI · KAFKA · WEBSOCKETS · POSTGRESQL · REDIS · PROMETHEUS

- Handled 1,000+ events/min with Kafka consumer groups, dead-letter routing, idempotency, Redis, and consumer-lag monitoring.

Ovia — Predictive Modelling Research (IEEE NELEX 2026)

PYTHON · XGBOOST · FASTAPI · POSTGRESQL · DOCKER

- Developed an XGBoost predictive modelling workflow with data processing, evaluation, API inference, and PostgreSQL; accepted as an IEEE conference paper.

AI RESEARCH & PUBLICATIONS

- 2025–present: Research on LLM inference output validation, hallucination detection, model failure analysis, and response reliability evaluation.
- 2026: IEEE NELEX conference paper — Ovia: XGBoost-Based Predictive Modelling for Women's Wellness Tracking.
- 2024: Journal publication — Fast Fashion's Impact on Youth & Sustainability, Asia Journal of Ethics and Sustainability.

EDUCATION

Jain (Deemed-to-be University) — *Bengaluru, India*
B.Tech — Computer Science & Business Systems | GPA: 8.3/10 — 2022 – 2026

Relevant coursework: Data Structures & Algorithms, DBMS, Operating Systems, Operations Research, Business Analytics, Computer Networks.

CERTIFICATIONS

Meta Backend Developer Professional Certificate (2024) · Apache Kafka — Confluent (2024) · Docker for Developers (2024)
GitHub Actions CI/CD (2024) · Google Cloud Fundamentals (2023) · Linux Command Line Essentials (2023) · IBM Machine Learning with Python (2023)

ACHIEVEMENTS & LEADERSHIP

- Top 5 finalist, Indegene Hackathon 2025 (2,000+ teams) — real-time anomaly detection backend.
- Open-source contributor to Backtrader and FastAPI-pagination (2024).
- Founder & Technical Lead, Echo Club — FinTech-focused student community, Jain University (2023–present).
- Founding Member & Tech/PR Lead, Tech Council SCSE — led technical and PR operations for three years.
- TCS IT Quiz — National Level Representative, Karnataka state selection (2023).

LANGUAGES

English (C1 professional) · Kannada (native) · Hindi (full proficiency)